Mohamed Medhat Kamal

Senior Software Engineer · Node.js · NestJS · TypeScript · Laravel | AI & LLM Integrations

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SUMMARY

Senior Software Engineer with 12+ years specializing in backend architecture, high-throughput APIs, and scalable distributed systems using Node.js, NestJS, TypeScript, and PHP/Laravel. Deep focus on core backend fundamentals: clean database modeling, query optimization, Redis caching strategies, asynchronous queues, and idempotent operations. Proven track record improving system performance on platforms serving 1M+ active users, cutting API response times by 60%, and maintaining 90%+ test coverage. Experienced in real-time telemetry (WebSockets, MQTT) and building practical production AI workflows with Claude Code, Anthropic API, and LangGraph.js.

SKILLS

Backend: Node.js (Express, NestJS), TypeScript, PHP 8+, Laravel 12+ (Octane/FrankenPHP, Horizon, Reverb, Sanctum), RESTful APIs (Swagger/OpenAPI), WebSockets, MQTT, multi-tenant architecture

AI & LLM: Claude API (Anthropic SDK), Ollama (local models), RAG (chunking, embeddings, pgvector + HNSW), tool calling & agent loops, LangChain.js, LangGraph.js, structured output with zod, guardrails & grounding checks, LLM evals, Claude Code

Frontend: React, TypeScript, Vue 3 (Composition API, Pinia, Vuex), Nuxt.js, Vuetify, Tailwind CSS, Bootstrap

Databases: MySQL, PostgreSQL, Redis, MongoDB

Testing & QA: Jest, Supertest, Vitest, PHPUnit, Mockery, TDD, integration testing, coverage-driven development

DevOps & Cloud: Docker, CI/CD, Linux, Nginx, AWS (EC2, RDS, S3, ELB), GCP, Northflank, DigitalOcean, Railway

Security & Auth: OAuth 2.0, JWT, Laravel Sanctum, RBAC, policies & gates, CASL

Observability & Analytics: Sentry, DataDog, Bugsnag, Laravel Telescope, Horizon, Uptime Kuma, UptimeRobot, Umami

Technical Ownership & Architecture: Architecture ownership, code reviews and development best practices, technical risk assessment, cross-functional alignment (Backend, Frontend, Mobile, QA, DevOps), translating business requirements into technical designs, working directly with CTOs and Founders, Agile/Scrum delivery

Domain expertise: Fintech & digital wallets (double-entry ledgers), mobility & car rental (1M+ users), industrial IoT & smart manufacturing (MES, OEE, ANDON), HR & workforce management, chatbots & messaging APIs, ERP integrations (Odoo), document automation

EXPERIENCE

Full-Stack Engineer (Contract) · Zimam (Imtithal Platform, 1M+ users) · Remote **Dec 2025 – Jun 2026**

- Built a dynamic DOCX-to-PDF document-generation pipeline (unoserver) powering four legal document types (Contract, NDA, Custody, Custody Handover) with template-driven variables, digital signatures with image cropping, Hijri date support, and caching.
- Designed and shipped the Application Invitations module end-to-end (migrations, API, mail, UI) for invite-based candidate registration.
- Implemented a unified hybrid storage service (local / Google Cloud Storage / AWS S3) with signed URLs, a secure file controller, and backward compatibility for legacy file paths.
- Built a two-way RMS/OMS platform integration: employee sync, role/credential propagation, and a dedicated sync command.
- Delivered a public landing page (Vue 3 + GSAP scroll animations) with a multi-step application form for Hajj/Umrah field-job candidates.
- Shipped custody management, work shifts, training programs/quizzes, and salary-objection modules; extended the Question Bank with bulk import and built a Cards Template Management System for dynamic ID/badge card generation.
- Partnered directly with the PM and GM on release planning and sprint breakdown, turning roadmap features into daily engineering tasks.
- Stabilized the legacy AngularJS dashboard, resolving defects across contract signing, PDF rendering, search, filters, validation, login flow, and responsive layout, and refactored the Authorizations and Training modules for maintainability and performance.

Stack: Node.js/TypeScript integrations, Laravel 11 / PHP 8.3, Vue 3 + TypeScript + Vuetify, AngularJS, MySQL, Redis, Docker, GCP, AWS S3

Full Stack & Embedded Systems Engineer (Contract) · SmartX Solutions (Industrial IoT) · Remote **Jul 2025 – Dec 2025**

- Developed a multi-tenant Smart Manufacturing SaaS platform built around four core modules: MES (production execution), CMMS (maintenance management), Quality management, and an automated Escalation engine, plus workforce management.
- Built a real-time IoT communication layer using WebSockets and MQTT for live factory-floor telemetry.
- Designed the IoT gateway device itself through the full hardware lifecycle: breadboard prototyping, C++ firmware development, testing, and iteration to production-ready devices connecting factory machines to the platform.
- Implemented OEE tracking, ANDON boards, shift management, and production analytics dashboards.
- Designed role-based access control across a multi-tenant architecture.
- Maintained 90%+ test coverage (PHPUnit unit + feature tests, Mockery) and integrated Sentry, Telescope, and Horizon for proactive monitoring.
- Ported the platform's escalation engine to NestJS.

Stack: NestJS, Node.js, TypeScript, PostgreSQL, Redis, MQTT, WebSockets, Docker, Jest, Sentry, Laravel 12

Back End Developer (Contract) · Telgani (car rental platform, KSA, 1M+ users) · Remote **Feb 2024 – Jul 2025**

- Designed and developed high-performance RESTful APIs (Laravel, MySQL, Redis) serving mobile applications with 1M+ active users.
- Co-architected a payments and wallet system (digital wallets, loyalty points, vouchers, a dynamic discount engine) using double-entry bookkeeping, rollback mechanisms, and financial audit trails with zero discrepancies.
- Optimized database queries and caching strategies, reducing API response time by 60%.
- Built the security layer: Laravel Sanctum bearer-token authentication plus authorization via policies, gates, and role-based permissions.
- Practiced TDD with comprehensive unit/feature/integration suites (PHPUnit, Mockery), maintaining 90%+ coverage and zero-regression deployments.
- Drove API-first development with full Swagger/OpenAPI documentation, enabling parallel frontend/backend work; refactored legacy code with SOLID principles, improving maintainability by 40%.
- Contributed to architectural decisions in system design meetings with the CTO and product team; collaborated cross-functionally with mobile, frontend, and QA teams on planning and task breakdown.
- Implemented observability with Telescope, Horizon, Bugsnag, and DataDog.

Stack: Node.js/TypeScript services, Laravel, MySQL, Redis, AWS (EC2, RDS, S3), Docker, WebSockets, Bugsnag, DataDog

Full-Stack Developer / Team Lead · Squadio (tech house, KSA) · Remote **Dec 2022 – Jan 2024**

- **Aklat:** joined as full-stack developer and became Team Lead as the team grew, owning technical direction and code review; resolved critical UI, security and performance issues; integrated Moyasar payments (STC Pay, Apple Pay, MADA) and Odoo ERP; improved subscription conversion by 35%.
- **Eat-Hub (mobile app):** architected the complete backend API from requirements to deployment; built the admin dashboard SPA (Vue.js); shipped real-time WebSocket chat between customers and restaurants.
- **Soiree Portal:** designed dual API systems (mobile + restaurant web app) with a real-time analytics dashboard and WebSocket notifications.

Stack: Node.js, Laravel, Vue.js, MySQL, Redis, WebSockets, Docker, AWS, Odoo API

Full-Stack Developer & Team Lead · Mobile World (chatbots, UAE) · Remote **Apr 2021 – Nov 2022**

- Led a team of 5 developers; introduced Agile Scrum, cutting the delivery cycle from 1 month to 1 week.
- Architected the Bot Builder Platform (low-code web app for generating chatbot databases) and delivered 15+ chatbot solutions across the MENA region.
- Built a reusable Vue.js component library for client dashboards, reducing build time by 50%; optimized chatbot responses to sub-200ms latency.

Early Career · Full-Stack Developer **Jun 2013 – Feb 2021**

- Delivered 10+ web applications across cosmetics ERP (SalyJolie), real estate (an Electron.js desktop accounting application for property managers with tenant and owner ledgers and 6+ accounting reports), car rental (Rentex), and B2B networking (Venue Manager, USA). Built RESTful APIs, Electron.js desktop applications, and Stripe payment integrations. Progressed from frontend to full-stack engineering.

EDUCATION

B.Sc. Information Technology (Web Development) · Advanced Academy, Giza, Egypt

LANGUAGES

Arabic (native) · English (professional)

LIVE PROJECTS

Open source: AI Real-Estate Assistant (bilingual RAG agent) · github.com/m-medhat-dev/real-estate-agent

www.imtithal47.com · www.telgani.com · phommunity.com

Video demos: youtube.com/playlist?list=PL-7vd1W-gHEb6ThtC9e9-Uxb6gO-f9xef